AI/ML RESEARCH METHODOLOGY BLUEPRINT

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Branch: BTech AI ML

Project Title / Topic: Neighbourhood-aware misbehavior detection for V2X networks (VeReMi Extension benchmark)

1. PROBLEM FORMULATION:

- Specific ML/AI Problem or Failure Mode Addressed:

Connected vehicles broadcast position, speed and heading; a misbehaving vehicle sends false data (spoofed position, frozen speed, copied trails) that misleads others. Most learned detectors judge each sender **in isolation**, so evidence from the surrounding vehicles goes unused, and it is untested how a neighbour-aware detector behaves when many neighbours are themselves malicious or when attackers collude and tell consistent lies. Failure mode: missed or slow detection of falsified data, and unknown robustness to malicious neighbours.

2. HYPOTHESIS FORMULATION:

- Null Hypothesis (H0):

Adding attention across nearby senders (neighbours) gives the same F1 as the same model without it on held-out traffic (F1 with neighbours = F1 without).

- Alternative Hypothesis (H1):

Neighbour attention gives higher F1 (F1 with > F1 without) with no increase in false positives. Secondary (H1b): the gain holds even when most of a sender's neighbours are attackers (H0b: the gain shrinks as malicious neighbours increase).

3. RESEARCH & EXPERIMENTAL DESIGN:

- Dataset to be used (e.g., ImageNet, Kaggle, Custom collected):

VeReMi Extension (public simulated benchmark, Zenodo 20090854; F2MD/VEINS/SUMO, Luxembourg scenario). Main study: 14 of its 19 attack types (RandomSpeedOffset archive excluded as mislabeled); 109,823 train / 29,547 validation / 68,971 test windows. Extension: 18 attacks, adding the 4 Sybil attacks (88,695 test windows).

- Baseline Model vs. Proposed Architecture/Modification:

Baselines: a speed-check rule; our first model (transformer on raw claims, no checks); the same model with neighbour attention switched off. **Proposed (v2):** small transformer (144k parameters) with temporal and cross-sender (neighbour) attention. Inputs: 9 claim values + 10 plausibility checks (kinematics, a road map learned from honest traffic, nearest other claim, exact self-repeat, copied trail, log movement). Training adds carefully injected lies (offset, fake stop); optional receiver-side memory.

- Evaluation Metrics (e.g., F1-Score, mAP, BLEU, AUC-ROC):

- **Precision:** of the senders we flag as malicious, the share that really are (few false alarms).
- **Recall:** of all malicious senders, the share we catch (few missed attackers).
- **F1-score:** balance of precision and recall (their harmonic mean); one-number summary of detection quality.
- **Accuracy:** share of all windows (honest and malicious) classified correctly.
- **False-positive rate (FPR):** share of honest senders wrongly flagged as malicious.
- **Detection latency:** seconds from an attacker's first appearance at a receiver to the first alarm on it.
- **Colluder recall at matched 1% FPR:** how many colluding attackers are caught when false alarms are held at 1%.

4. PROPOSED VALIDATION STRATEGY:

- (e.g., K-Fold Cross Validation, Statistical Significance Test):

- Train on the first simulated hour; test on the second hour, whose receivers are split into separate validation (30%) and test (70%) blocks; thresholds and memory decay chosen on validation only; 2 seeds per model.
- Two label views: **message labels** (attacker counted only in windows where it sent a falsified message) and **vehicle labels** (every window of an attacker vehicle).
- Paired bootstrap over receivers (95% CI) with Holm correction across attacks; results by malicious-neighbour share; collusion stress test with consistent lies (offset, fake stop, spoofed convoy, on-road lies).

5. EXPECTED AI / COMPUTATIONAL OUTCOME:

- **Results (14 attacks, 68,971 test windows, 2 seeds).** Message labels: v2 precision 98.65%, recall 97.12%, F1 97.88%, accuracy 98.69%, FPR 0.60%; with memory P 98.16%, R 97.78%, F1 97.97%, accuracy 98.74%. Vehicle labels: v2 F1 97.01% (recall 95.41%); with memory F1 97.12%. First model F1 94.9% (vehicle labels); speed-check rule F1 78.9%.
- **Neighbour attention:** +0.3 F1 points (95% CI about [0.2, 0.4]), significant on 8-10 of 14 attacks, FPR 0.90% to 0.60%, so H0 is rejected. H1b is not supported: the gain shrinks and turns slightly negative when attackers are the majority of neighbours.
- **Latency:** mean 0.9 s to first alarm (median 0); 96% of attacker-receiver pairs flagged. **Collusion:** v2 catches 50 m offsets (recall 0.83-0.86), fake stops (about 0.70) and unseen spoofed convoys (0.87-0.90), where the first model caught at most 0.04; lies that follow the road map stay hard (0.23-0.48).
- **Sybil extension (18 attacks, senders tracked by pseudonym, 3-model ensemble):** message labels P 96.30%, R 95.46%, F1 95.88%, accuracy 97.04%, FPR 2.07%; vehicle labels F1 95.38%. Neighbour attention adds about +1.3 F1 here. Main miss: GridSybil ghosts (recall 0.82), which pass every content check.
- Reference: a 2025 IEEE T-ITS survey lists 97.37% for VeReMi (original 5 attacks, no Sybil); on 14 attacks v2 + memory is above it on all four metrics with message labels, on 18 attacks all four are below it; not like-for-like. Open: original VeReMi/NextGen check, dense traffic, a 3rd seed; on-road lies and GridSybil ghosts need an independent signal (signal strength, sensors, RSUs).